

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>  
To determine the immune response elicited by the pathogen causing the patient's condition, we need to consider the characteristics of the pathogen and the symptoms presented. The patient has been diagnosed with community-acquired pneumonia, which is often caused by bacteria such as Streptococcus pneumoniae. This type of pneumonia is typically caused by a gram-positive bacterium that is commonly found in the environment and can be transmitted through respiratory droplets.

The symptoms presented, such as sore throat, non-productive cough, fever, and patchy reticular opacities in the perihilar regions of both lungs, are consistent with a bacterial infection. The treatment with cephalexin, a beta-lactam antibiotic, indicates that the pathogen is likely a gram-positive bacterium.

Considering the characteristics of gram-positive bacteria, we can infer that the immune response elicited by the pathogen is likely to be a cell-mediated response, which involves the activation of T cells, particularly Th1 cells. Th1 cells play a crucial role in defending against intracellular pathogens, including gram-positive bacteria.

Therefore, the immune response elicited by the pathogen is primarily an Th1-cell response, which involves the activation of T cells to produce cytokines that activate macrophages and other immune cells to eliminate the infection.

</think><answer> B. It primarily induces the Th1-cell response.